CH5 Special Discrete Distributions

Bernoulli and Binomial R.V.

Bernoulli Definition

X is a Bernoulli trial (or Bernoulli random variable) with parameter p if sample space S={s, f}, where s is called a success and f a failure and X(s)=1, X(f)=0.

The probability mass function of X is

$$p(x) = \begin{cases} p & x = 1 \\ 1-p \equiv q & x = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = 0 \times P(X=0) + 1 \times P(X=1) = p$$

$$E(X^2) = 0^2 \times P(X=0) + 1^2 \times P(X=1) = p$$

$$Var(X) = E(X^2) – (E(X))^2 = p – p^2 = p(1-p)$$

Binomial R.V. Definition

$X \sim B(n,p)$ X: The number of success

n: Perform n independently event

p: The probability of success

The p.m.f. of $X \sim B(n,p)$

$$p(x) = P(X=x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{if } x = 0,1,2,\ldots,n \\ 0 & \text{elsewhere.} \end{cases}$$

$$E(X) = np$$

$$E(X^2) = n^2p^2 - np^2 + np$$

If X is a binomial random varibale with parameters B(n,p) , we can obtain

$$\sum_{x=0}^{n} p(x) = \sum_{x=0}^{n} \binom{n}{x} p^x (1-p)^{n-x} = [p + (1-p)]^n = 1^n = 1$$

Note: Multinomial theorem is used

Theorem

If X is a binomial random variable with parameter B(n,p), then as k goes from 0 to n, P(X=k) first increases monotonically and then decreases monotonically, reaching its maximum value when k is the largest integer less than or equal to (n+1)p.

假設我們進行 n=100 次試驗，每次成功的機率 p=0.665。 $(n+1)p = 101 \times 0.665 = 67.165$

根據定義，k 是小於或等於 67.165 的最大整數，所以 k=67

Ex 5.2 A restaurant serves 8 entrees of fish, 12 of beef, and 10 of poultry. If customers select from these entrees randomly, what is the probability that 2 of the next four customers order fish entrees?

$X(4, \frac{8}{30})$

$P(X=2) = \binom{4}{2} \left(\frac{8}{30}\right)^2 \left(\frac{22}{30}\right)^{4-2}$

Ex 5.2 The probability of a bit error in a communication line is 0.001. Find the probability that a block of 10000 bits contains two or more errors?

$X(10000, 0.001)$

$1 - P(X=0) - P(X=1)$

Ex 5.2 Consider a jury trial in which it takes 8 of the 12 jurors to convict; that is, in order for the defendant to be convicted, at least 8 of the jurors must vote him guilty. Assume that each of the jurors makes the right decision with probability p ,while g represents the probability that the defendant is guilty. What is the probability that the jury makes a right decision?

情況 1: (有罪) → 至少 8 人做出正確判決

$P(A g) = P(A|g) \cdot P(g) = \sum_{k=8}^{12} C_k^{12} p^k (1-p)^{12-k} \cdot P(g)$

情況 2: (無罪) → 至少 5 人判無罪

$P(A|g^c) = P(A|g^c) \cdot P(g^c) = \sum_{k=5}^{12} C_k^{12} p^k (1-p)^{12-k} \cdot P(g)$

Mean of Binomial random variable

If X is a binomial random variable

$E(X) = np$

$E[X(X-1)] = n^2p^2 - np^2$

$E(X^2) = n^2p^2 - np^2 + np$

$Var(X) = E(X^2) - (E(X))^2 = np(1-p)$

5.2 Poisson random variable

Definition

When $n \to \infty, p(x) \to 0$ in a binomial probability function, then $n \times p = \lambda$

A discrete random variable X with possible values 0, 1, 2, 3, … is called Poisso

with parameter λ , if $P(X = i) \to \frac{e^{-\lambda}\lambda^i}{i!}$

$E(X) = \lambda$

$Var(X) = \lambda$

$E(X^2) = \lambda^2 + \lambda$

This can approximate the situation that happen at the very low chance but large

amount. If so the λ is the mean value of it.

Poisson random variable: Example

- **Ex 5.11 Suppose that, on average, in every three pages of a book there is one typographical error. If the number of typographical errors on a single page of the book is a Poisson random variable, what is the probability of at least one error on a specific page of the book?**

- **Sol: E(X)=1/3 so X~P(1/3)**

- $$P(X = n) = \frac{(1/3)^n e^{-1/3}}{n!}$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-1/3} \approx 0.28$$

5.3 Other Discrete Random Variables

Geometric Random Variable

Do n times in an experiment while results is $\{\text{failed} \times (n-1), \text{success}\}$

$X \sim Geo(p) = P(X=n) = (1-p)^{n-1} \times p, \quad n = 1, 2, 3$

$E(X) = \dfrac{1}{p}$

$E(X^2) = \dfrac{2-p}{p^2}$

$Var(X) = \dfrac{1-p}{p^2}$

Because the p.m.f of the Geometric Random Variable is like 等比數列 (r=1-p) 故得名

Exercies p33. TODo

Theorem (Memoryless Property)

$P(X \geq n+m | X > m) = P(X \geq n)$

Negative Binomial R.V.

Let X be the number of the experiments until the r^{th} success,

which contains k failure, the probability of success p is:

實驗次數：$n = k + r$

$P(X=n) = C^{n-1}_{r-1} \cdot p^r \cdot (1-p)^k$ denoted: $X \sim NB(r, p)$

$E(X) = \dfrac{r}{p}$

Hypergeometric R.V.

N: Total item count

D: Defectived item count

n: Drawn item count which less than D or N-D $n \leq \min(D, N-D)$

X: be the number of defective items drawn

$P(X=x) = \dfrac{C^D_x C^{N-D}_{n-x}}{C^N_n}$ for x in 0 ~ n

$E(X) = n \cdot \dfrac{D}{N}$

Which denoted: $X \sim \text{HGeo}(N, D, n)$

Summarize

Bernoulli R.V. (Bernoulli Trial)

實驗只有成功或失敗

$$P(X=k)=\begin{cases} p & k=1 \\ 1-p\equiv q & k=0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X)=0\times P(X=0)+1\times P(X=1)=p$$

$$E(X^2)=0^2\times P(X=0)+1^2\times P(X=1)=p$$

$$Var(X)=E(X^2)–(E(X))^2=p–p^2=p(1-p)$$

Binomial R.V. $X\sim B(n,p)$

實驗 N 次，有 r 次的成功的機率

$$P(X=x)=\begin{cases} \binom{n}{x}p^x(1-p)^{n-x} & \text{if } x=0,1,2,\ldots,n \\ 0 & \text{elsewhere.} \end{cases}$$

$$E(X)=np$$

$$E(X^2)=p^2n^2-p^2n+pn$$

$$Var(X)=E(X^2)-(E(X))^2=np(1-p)$$

Negative Binomial R.V. $X\sim NB(r,p)$

前面都失敗，但是最後發生第 r 次成功

$$P(X=n)=C^{n-1}_{r-1}\cdot p^r\cdot(1-p)^k$$

$$E(X)=\frac{r}{p}$$

Poisson R.V. $X\sim P(\lambda)$

估計極小機率在大量實驗的機率

$$P(X=i)\rightarrow\frac{e^{-\lambda}\lambda^i}{i!}\qquad\lambda=np$$

$$E(X)=\lambda$$

$$E(X^2)=\lambda^2+\lambda$$

$$Var(X)=\lambda$$

Geometric R.V. $X\sim Geo(p)$

前面都失敗，第 n 次才迎來首次成功

$$P(X=n)=(1-p)^{n-1}\times p,\quad n=1,2,3$$

$$E(X)=\frac{1}{p}$$

$$E(X^2)=\frac{2-p}{p^2}$$

$$Var(X)=\frac{1-p}{p^2}$$

Hypergeometric R.V. $X\sim \mathrm{HGeo}(N,D,n)$

N個物件標記D個，然後抽出n個物件有x個標記的機率

$$P(X=x)=\frac{C^D_xC^{N-D}_{n-x}}{C^N_n}\qquad x\in[0,n]$$

$$E(X)=n\cdot\frac{D}{N}$$